

AI Nexus Intake Engine

Architectural Blueprint for Scalable Clinical Ingestion and Verification

1. Executive Summary & Operational Bounds

The AI Nexus Intake Engine acts as the definitive data translation boundary of our clinic infrastructure. Its primary directive is safely normalizing erratic patient reporting into predictable, secure formats. To achieve infrastructural stability within this phase, the ingestion parameters are strictly bounded to maintain highly performant computing loops.

Ingested Baseline Records: 20K+ Historical Units	Maximum Daily Updates: < 10,000 Scheduled Requests
Active Journals Indexed: 140K+ Medical Profiles	Global Target Jurisdictions: 20+ Cross-Border Regions

2. Algorithmic Quality & Statistical Verification

Deploying large language model engines within a live clinical workflow demands rigorous guardrails to combat clinical omissions. The core ingestion framework targets optimization parameters across an evaluation baseline of 10,000 verified PubMed citations, prioritizing the mitigation of false negatives.

2.1 Statistical Validation Targets

- **False Negative Rate (FNR): < 5%** — The critical safety threshold. Missed indications of critical findings are strictly mitigated to avoid diagnostic delays.
- **False Positive Rate (FPR): < 10%** — Calibrated carefully to protect medical staff from systemic alert fatigue, preventing routine complaints from triggering critical queues.

- **Screening Time Optimization: Up to 80% Reduction** — Streamlining patient background analysis by immediately parsing complaints into structured EHR payloads.

Validation Method: The False Negative Rate is actively audited against programmatic benchmarks using:

$$FNR = FN / (TP + FN) < 0.05$$

Where TP indicates True Positives correctly captured by the intake pipeline, and FN denotes clinical indicators that bypassed model extraction.

3. Scalable Ingestion Control & Rate Limiting

To strictly enforce the operational mandate of fewer than 10,000 daily transaction updates, the intake API infrastructure utilizes a token-bucket strategy implemented directly at the gateway layer.

The system computes a uniform transaction flow rate across a rolling 24-hour window, yielding a baseline allowance of roughly 416 updates per hour. Payloads exceeding this threshold are systematically moved into an asynchronous Redis-backed staging queue, shielding downstream databases from performance spikes and preserving query execution speeds.

4. Multi-Layer Execution Pipeline

The intake infrastructure implements a decoupled three-stage operational cycle:

1. **Conversational Interface Layer:** Captures unstructured patient transcripts or text inputs cleanly via responsive front-end forms.
2. **Heuristic Extraction Layer (NER):** Maps variable clinical syntax to rigorous medical database entities, assigning explicit SNOMED-CT findings and matching ICD-10 diagnostic entries.

3. **Deterministic Safety Gate:** A non-heuristic code path evaluating inputs for emergency indicators. High-priority terms instantly bypass clinical AI queues to trigger urgent care pathways.

Critical Safety Override: The safety gate relies on explicit keyword matrices and strict deterministic pattern matches. If an immediate acute symptom is recognized, the pipeline drops the standard multi-turn intake sequence and immediately defaults to the urgent triaging state.